

IGNASHINO, Russian Federation

WMO: 306860

Lat: 53.467N

Lon: 122.400E

Elev: 967

StdP: 14.19

Time Zone: 9.00 (E09)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.6	-38.7	-46.8	0.4	-42.0	-43.0	0.5	-38.4	17.2	-2.8	15.6	-4.0	0.8	0	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.6	87.6	68.1	83.8	66.8	80.1	65.6	71.1	81.6	69.3	79.0	67.6	76.4	3.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.9	106.2	75.6	66.2	99.9	73.5	64.5	94.2	72.2	35.8	82.0	34.2	79.0	32.8	76.2	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	13.9	11.5	-46.4	93.4	3.6	4.1	-49.0	96.3	-51.0	98.7	-53.0	101.0	-55.6	104.0		
(5)			-46.3	75.1	3.2	2.9	-48.6	77.2	-50.5	78.8	-52.3	80.5	-54.6	82.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.3	-18.6	-9.1	10.0	33.1	48.8	62.0	67.0	62.3	48.1	28.0	-0.3	-17.5		
	DBStd	32.19	9.42	9.41	12.24	8.59	6.65	6.14	5.16	6.23	7.79	9.96	13.84	10.16		
	HDD50	10047	2127	1654	1240	510	102	2	0	3	126	683	1508	2092		
	HDD65	14293	2592	2074	1705	958	505	127	39	123	507	1148	1958	2557		
	CDD50	1410	0	0	0	2	63	363	528	385	69	0	0	0		
	CDD65	182	0	0	0	0	1	38	102	41	0	0	0	0		
	CDH74	2563	0	0	0	2	64	826	1149	504	18	0	0	0		
	CDH80	810	0	0	0	0	10	269	397	134	0	0	0	0		
	WSAvg	3.6	2.8	2.9	3.9	4.8	5.0	3.7	3.1	3.4	3.9	4.1	3.0	2.7		
	PrecAvg	16.6	0.1	0.2	0.3	0.8	1.3	2.5	4.1	4.0	1.9	0.7	0.4	0.2		
(15) Precipitation	PrecMax	25.0	0.4	0.4	0.7	1.7	2.7	7.0	8.7	9.9	4.7	2.1	0.9	0.5		
	PrecMin	10.8	0.0	0.0	0.0	0.0	0.2	0.6	1.5	1.3	0.2	0.0	0.1	0.0		
	PrecStd	3.5	0.1	0.1	0.2	0.4	0.7	1.6	1.6	2.1	0.9	0.5	0.2	0.1		
	PrecStd	3.5	0.1	0.1	0.2	0.4	0.7	1.6	1.6	2.1	0.9	0.5	0.2	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	27.1	47.7	68.1	81.0	91.6	93.6	88.7	76.8	59.2	33.9	15.3		
		MCWB	8.2	22.1	36.7	48.8	58.1	69.1	69.4	69.4	61.7	46.5	29.0	13.7		
	2%	DB	6.3	18.4	41.3	60.4	75.0	87.0	88.2	83.7	71.7	53.1	28.9	9.8		
		MCWB	4.5	15.3	32.3	45.2	54.4	66.9	68.2	68.0	59.2	42.7	25.9	7.8		
	5%	DB	2.0	13.2	36.0	54.8	70.4	82.5	84.7	79.8	67.2	48.0	25.1	4.3		
		MCWB	0.6	10.5	29.1	41.8	52.5	64.3	68.1	66.8	56.6	39.6	22.7	2.9		
	10%	DB	-2.3	8.6	31.6	49.7	65.7	78.6	80.6	75.7	62.8	43.4	20.6	-1.3		
		MCWB	-3.4	6.6	26.3	38.4	50.5	62.7	67.1	64.6	53.2	36.6	18.5	-2.3		
	0.4%	WB	8.2	22.2	37.0	50.4	60.2	72.9	73.7	73.0	64.6	47.5	29.5	13.9		
		MCDB	10.3	26.8	46.6	66.5	75.4	86.9	83.3	84.1	73.9	56.9	32.6	15.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	4.7	15.5	32.8	45.9	56.8	69.0	71.6	70.2	60.9	43.7	26.2	7.9		
		MCDB	6.4	18.3	40.4	59.5	69.8	81.8	82.0	79.6	68.5	51.6	28.6	9.8		
	5%	WB	0.7	10.6	29.6	42.3	54.2	66.7	70.0	68.2	58.4	40.4	22.8	2.9		
		MCDB	2.0	13.1	35.5	53.3	66.7	79.0	79.7	76.4	64.1	46.8	25.1	4.3		
	10%	WB	-3.3	6.6	26.5	39.3	52.0	64.4	68.5	66.2	55.4	37.0	18.6	-2.2		
		MCDB	-2.3	8.8	31.6	48.9	63.8	75.9	77.2	73.0	61.7	43.2	20.7	-1.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.9	31.5	33.3	26.2	27.4	28.0	23.6	23.3	25.0	23.4	23.1	20.0		
		MCDBR	24.2	33.1	32.8	35.4	38.9	37.1	33.9	31.0	31.5	29.2	22.9	21.0		
		MCWBR	22.7	29.6	24.7	21.2	20.6	17.9	15.5	15.9	19.0	19.4	19.5	19.6		
		MCDBR	24.1	32.8	31.1	32.3	32.3	31.4	25.1	25.4	26.0	28.4	22.5	21.0		
	5% WB	MCWBR	22.6	29.4	23.6	19.9	18.3	16.4	13.1	14.2	16.9	19.5	19.1	19.6		
		MCWBR	22.6	29.4	23.6	19.9	18.3	16.4	13.1	14.2	16.9	19.5	19.1	19.6		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.287	0.336	0.466	0.536	0.489	0.462	0.419	0.366	0.334	0.290	0.264		
	taud		2.130	2.093	2.033	1.856	1.803	2.040	2.175	2.280	2.364	2.302	2.149	2.057		
	Ebn at Noon		224	253	263	239	229	244	249	254	255	235	205	189		
	Edh at Noon		21	31	41	57	63	50	43	37	30	25	21	19		
(44) All-Sky Solar Radiation	RadAvg		398	763	1297	1661	1664	1910	1662	1456	1107	724	429	297		
	RadStd		16	20	50	113	121	165	146	147	91	38	20	15		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page